Week 7 Project Progress Report

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Course Name: ECE501 Digital Image Processing (DIP)
Group Number: Group 15
Group Members: Rinal Shah, Varun Hotani, Aadi Mehta, Pratik Agrawal
Week Number: Week 7
Date: October 31, 2026
Deliverable: Image-level detection with a calibrated confidence score; cross-validated in-domain reference against the AFID capture shift; 64 → 32 refinement cascade; small-defect protocol study; frozen evaluation protocol
Branch: `feat/week-7` (merged into `main`)

Objectives of the Week

The Week 6 report ended with four items for this week. Each is addressed here:

- **Image-level detection and confidence.** Hold out defect-free strips as true negatives, decide per strip whether it is defective, and attach a confidence with a provable false-alarm guarantee.
- **Capture shift.** Use the defect-free *regions* of defective strips as extra reference data. Cross-validate by strip, so that no strip is ever scored by a model that saw any of its pixels.
- **Small defects.** Add a 32×32 level to the refinement cascade, and measure how the 25-pixel patch-label rule hides the smallest defects.
- **End-semester preparation.** Freeze the evaluation protocol and make every reported table regenerate from committed results.

Summary of the Week

Item	Outcome
Protocol	Per fabric, the 20 defect-free strips split by image (seeded): 12 fit, 4 calibrate, 4 held-out negatives. 105 defective + 28 defect-free test strips
Image level	Accuracy **68.42 %**, detection rate 60.00 %, **zero false alarms** on the 28 negatives ($\alpha = 0.05$), AUROC **0.790**
Localization	Unchanged by the new protocol: patch F1 18.38 % (pooled 23.02 %), pixel IoU 1.36 %, pixel recall 48.41 %
Threshold-free diagnosis	New per-strip *window AUROC*: 0.536 for the default, 0.609 with the in-domain reference. The descriptors rank defect windows only slightly better than chance within a strip.
In-domain reference	Raises window AUROC (broken ends 0.12 → 0.69) but over-tightens τ: pixel recall 48 % → 11 %. Kept as an option, not the default.
Small defects	64 → 32 cascade: pixel IoU 1.36 → 1.52 %. "Any defect pixel" labels: warp balls measurable for the first time (window AUROC 0.826)
Quality	86 tests (72 last week); every table in this report is generated from `Results/` by `reports/week-7/make_tables.py`

Work Completed This Week

A. A Frozen, Three-Way Evaluation Protocol

Until Week 6 every test strip was defective. That measures localization, but it cannot measure what an inspection line needs first: *whether a roll segment should be stopped at all.* Each fabric's 20 defect-free strips are now split *by strip* with a seeded permutation into three disjoint sets:

$$\underbrace{12}_{\text{fit } \boldsymbol{\mu}_f, \boldsymbol{\Sigma}_f} + \underbrace{4}_{\text{calibrate } \tau_f, \text{ confidence}} + \underbrace{4}_{\text{image-level negatives}} \tag{1}$$

(`calibration_fraction` = `test_fraction` = 0.2). The test set is therefore 105 defective strips (every one with a mask and a reference fabric) plus 28 defect-free strips, none of which touches fitting or calibration. Patch and pixel metrics are still computed on defective strips only; image-level metrics use all 133. Moving from 15 to 12 fit strips changed the localization results by less than 0.3 percentage points (patch F1 18.14 → 18.38 %, IoU 1.36 % both weeks).

B. Image-Level Decision and Conformal Confidence

Image score. A strip is as anomalous as its most anomalous window, measured in units of its fabric's threshold:

$$S = \frac{\max_i s_i}{\tau_f} \qquad (S \geq 1 \iff \text{at least one window is flagged}). \tag{2}$$

Dividing by τ_f puts all seven fabrics on one scale, so one calibration set can serve all of them.

Confidence. We use a conformal p-value [1, 2]. Let $S_1, \ldots, S_n$ be the image scores of the $n = 28$ held-out *calibration* strips (4 per fabric, never fitted on). For a test strip with score S,

$$p(S) = \frac{1 + \#\{j : S_j \geq S\}}{n + 1}, \qquad \text{confidence} = 1 - p(S), \qquad \text{defective} \iff p(S) \leq \alpha. \tag{3}$$

If defect-free test strips are exchangeable with the calibration strips (same capture, same distribution), then $\Pr[p \leq \alpha] \leq \alpha$. The false-alarm rate is bounded by α without any assumption on the score distribution. With $n = 28$ the smallest attainable p-value is $1/29 \approx 0.034$, so $\alpha = 0.05$ means "more anomalous than every calibration strip". A unit test checks the bound empirically: 5,000 exchangeable draws against 199 calibration scores give a flagged rate of at most 6 % at $\alpha = 0.05$.

C. Cross-Validated In-Domain Reference

Week 6 showed that normal windows of defective strips score far above held-out *defect-free* strips for five fabrics, a capture shift between AFID's two image sets. The natural fix is to learn "normal" from the same capture: the windows of defective strips that contain *no* defect pixel. Doing this naively leaks test information, so `model.reference = "in_domain"` runs a strip-level cross-validation:

1. Per fabric, the defective strips are shuffled (seeded) and dealt into $K = \min(5, n_f)$ folds; that fabric's negatives are dealt round-robin into the same folds.
2. For fold j, the training strips are the fabric's defective strips *outside* fold j. They are split again by strip into an inner fit part and an inner calibration part (20 %).
3. The reference is fitted on the defect-free fit strips *plus* the zero-defect windows of the inner-fit training strips. τ_f and the conformal calibration use the defect-free calibration strips *plus* the zero-defect windows of the inner-calibration strips. This is done at 256×256 and at every refinement level.
4. Fold j's strips are then scored. Every strip is scored exactly once, by a model that never saw any of its windows.

A window counts as defect-free when its summed-area-table count of mask pixels is exactly 0. Fabric 05 has a single defective strip, so there is nothing to learn from; it keeps the defect-free reference. The fold assignment, the inner split and the background selection are covered by unit tests.

D. Refinement Cascade

`refine.sizes` replaces Week 6's single fine level with a coarse-to-fine list, e.g. $[64, 32]$ (stride = size/2). The windows flagged at one level bound the next:

$$R_0 = \bigcup_{w \in \mathcal{F}_{256}} w, \qquad R_\ell = \bigcup \{ v \in \mathcal{V}_{s_\ell} : v \subseteq R_{\ell-1},\ d_f^{(s_\ell)}(\mathbf{x}_v) \geq \tau_f^{(s_\ell)} \}, \qquad \hat{M} = \mathrm{morph}(R_L), \quad (4)$$

with one Mahalanobis reference and one held-out τ per fabric and level. The 32×32 level needs about 3,700 windows per strip, so it was added to the feature cache once (about 12 minutes for all 246 strips on 8 processes); every later run reuses it.

E. Threshold-Free Window AUROC

All earlier patch metrics depend on τ. To separate "the descriptor cannot tell defect from background" from "the threshold is in the wrong place", each defective strip now also reports the AUROC of its window scores against its window labels:

$$\mathrm{AUROC}_{\mathrm{strip}} = \Pr\left[s_i > s_j \mid i \text{ defect window},\ j \text{ normal window of the same strip} \right], \quad (5)$$

averaged over strips that have both kinds of window. 0.5 is chance; values below 0.5 mean the defect scores *lower* than its surroundings.

Quantitative Results and Analysis

All numbers are on **real AFID data**, with seed 0 and the protocol of Eq. (1): 105 defective strips for patch and pixel metrics, plus 28 defect-free strips for image-level metrics. Tables 1–4 are generated from `Results/final/` and `Results/ablation/`.

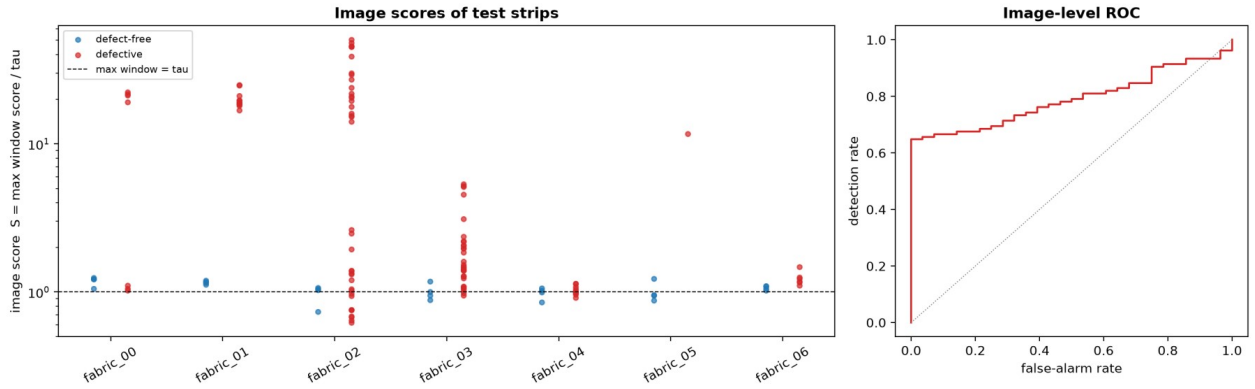


Figure 1: Left: image scores S (Eq. 2, log scale) of the 28 held-out defect-free strips (blue) and the 105 defective strips (red), per fabric; the dashed line is $S = 1$. Right: image-level ROC over all 133 strips (AUROC 0.790).

Table 1: Ablation (one factor at a time around the default; %, except AUROC). Patch/pixel columns: 105 defective strips. Image columns: the same plus 28 defect-free strips.

	Localization (defective strips)					Image level			
Variant	**Patch F1**	**Pooled F1**	**Win. AUROC**	**IoU**	**Pix. rec.**	**Acc.**	**Det. rate**	**Spec.**	**AUROC**
default	18.38	23.02	0.536	1.36	48.41	68.42	60.00	100.00	0.790
features: glcm	18.18	22.59	0.529	1.38	49.10	69.17	60.95	100.00	0.792
features: lbp	12.76	21.74	0.458	1.13	39.58	63.16	54.29	96.43	0.796
features: gabor	15.81	22.69	0.529	0.89	56.58	72.93	65.71	100.00	0.878
features: glcm+lbp+gabor	16.49	22.64	0.540	1.05	57.73	70.68	62.86	100.00	0.853
scoring: strip mahalanobis	8.16	12.34	0.546	0.46	7.01	33.83	17.14	96.43	0.366
scoring: hybrid (week 5)	14.01	14.85	0.574	0.71	26.76	49.62	36.19	100.00	0.675
scoring: relative	14.57	16.57	0.551	0.87	29.99	48.12	34.29	100.00	0.689
baseline: one-class svm	16.71	23.43	0.572	1.32	43.70	61.65	55.24	85.71	0.697
baseline: isolation forest	14.40	18.68	0.616	0.95	40.13	47.37	33.33	100.00	0.669
reference: global (fixed)	5.77	12.02	0.461	0.08	22.82	42.11	26.67	100.00	0.744
reference: in-domain (cross-validated)	7.86	12.54	0.609	0.35	10.60	45.11	30.48	100.00	0.736
k = 2.0	18.82	23.23	0.536	1.41	50.41	69.17	60.95	100.00	0.789
k = 3.0	18.60	22.40	0.536	1.37	46.98	68.42	60.00	100.00	0.792
refine: off	18.38	23.02	0.536	0.71	54.93	68.42	60.00	100.00	0.790
refine: 64 then 32	18.38	23.02	0.536	1.52	43.75	68.42	60.00	100.00	0.790
refine: 32 only	18.38	23.02	0.536	1.55	46.10	68.42	60.00	100.00	0.790
patch label: any defect pixel	19.77	24.52	0.557	1.36	48.41	68.42	60.00	100.00	0.790
image alpha = 0.10	18.38	23.02	0.536	1.36	48.41	69.92	61.90	100.00	0.790
week 5 setting	14.01	14.85	0.574	0.30	32.88	49.62	36.19	100.00	0.675

Table 2: Image-level detection per fabric (default, $\alpha = 0.05$). Negatives: 4 held-out defect-free strips per fabric.

Fabric	**Defective**	**Defect-free**	**Accuracy**	**Detection rate**	**Specificity**	**AUROC**
fabric_00	7	4	72.73	57.14	100.00	0.643
fabric_01	12	4	100.00	100.00	100.00	1.000
fabric_02	38	4	66.67	63.16	100.00	0.770
fabric_03	31	4	71.43	67.74	100.00	0.927
fabric_04	10	4	28.57	0.00	100.00	0.550
fabric_05	1	4	100.00	100.00	100.00	1.000
fabric_06	6	4	50.00	16.67	100.00	1.000

Table 3: Default configuration per fabric (defective strips; %, except window AUROC). "Detected" is the image-level detection rate.

Fabric	n	**Patch prec.**	**Patch rec.**	**Patch F1**	**Win. AUROC**	**Pixel IoU**	**Pixel rec.**	**Detected**
fabric_00	7	7.70	71.43	13.68	0.461	0.06	57.14	57.14
fabric_01	12	14.83	100.00	21.05	0.366	1.93	89.15	100.00
fabric_02	38	3.92	75.00	7.20	0.483	0.09	55.26	63.16
fabric_03	31	38.39	62.31	33.57	0.633	3.73	39.15	67.74
fabric_04	10	0.00	0.00	0.00	0.640	0.00	0.00	0.00
fabric_05	1	6.90	100.00	12.90	0.074	0.03	100.00	100.00
fabric_06	6	25.00	30.00	25.00	0.759	0.08	31.67	16.67

Table 4: Default configuration per official defect class (defective strips; %, except window AUROC). "–": undefined (no window holds more than 25 mask pixels).

Defect	n	**Patch prec.**	**Patch rec.**	**Patch F1**	**Win. AUROC**	**Pixel IoU**	**Pixel rec.**	**Detected**
broken_end	9	6.53	88.89	12.14	0.121	0.11	85.66	100.00
broken_pick	10	76.00	60.60	52.77	0.776	10.91	47.99	80.00
broken_yarn	8	6.91	87.50	11.30	0.278	0.12	87.50	87.50
contamination	1	0.00	0.00	0.00	0.793	0.00	0.00	0.00
crease	5	7.61	60.00	13.25	0.546	0.49	80.00	60.00
cut_selvage	9	41.93	55.56	40.42	0.864	0.22	43.34	33.33
fuzzy_ball	39	7.61	47.22	9.25	0.696	0.01	25.64	43.59
knot	1	6.90	100.00	12.90	0.074	0.03	100.00	100.00
nep	14	4.75	69.23	8.82	0.442	0.02	64.29	85.71
warp_ball	5	0.00	–	0.00	–	0.02	20.00	0.00
weft_crack	1	93.55	100.00	96.67	0.328	22.39	98.83	100.00
weft_curling	3	14.84	50.00	22.88	0.488	1.57	47.53	66.67

Discussion of Findings

1. **The conformal decision is what makes image-level detection usable.** With the naive rule "defective if any window crosses τ_f" ($S \geq 1$), 20 of the 28 defect-free test strips would be false alarms, because a 4,096-px strip has about 30 windows and each may cross a per-window threshold by chance. The conformal rule (Eq. 3) instead calibrates the *strip-level* statistic. Its effective cut-off falls between $S = 1.34$ and $S = 1.38$. It produced no false alarm on the 28 negatives while detecting 63 of 105 defective strips (60.0 %), and the threshold-free AUROC is 0.790.

2. **Image-level numbers are inflated where AFID's captures differ.** The median image score of defective strips is 8.4–19.2 for fabrics 00, 01, 02 and 05, against 0.95–1.22 for their defect-free test strips (Figure 1). That gap is mostly the capture shift reported in Week 6, not the defect. The fair test is fabrics 03, 04 and 06, whose captures transfer:

   - **Fabric 03:** AUROC 0.927 and 67.7 % detection, a real defect signal (broken picks, weft curling).
   - **Fabric 06:** AUROC 1.000 but only 16.7 % detection.
   - **Fabric 04:** its 10 small defects are not separable at all (AUROC 0.550, nothing detected).

3. **Within a strip, window scores barely rank defects above background.** The mean window AUROC of the default is 0.536 (Table 1). For broken ends and knots it is far *below* 0.5: in those shifted fabrics the whole strip is anomalous, and the thin defect window is no more anomalous than its neighbours. This threshold-free number explains why every thresholded variant of the last three weeks stayed below about 20 % patch precision. The bottleneck is the separability of 256×256 GLCM+LBP descriptors, not the choice of k, scorer or reference.

4. **In-domain background fixes the ranking but not the threshold.** Adding other strips' defect-free windows raises window AUROC from 0.536 to 0.609, the largest gain of any variant (broken ends $0.12 \rightarrow 0.69$). Thresholded results nevertheless fall: patch F1 7.86 %, pixel recall 10.6 %. The mixed reference has a much broader covariance, and its calibration windows include the most heterogeneous background, so τ_f rises until most defects pass under it. The shift is therefore not a single offset: defective strips also differ *among themselves* about as much as the defects differ from them. A model that learns the right ranking needs a threshold set by defect-aware validation, which AFID's sizes per fabric (1–38 defective strips) barely allow.

5. **Adaptivity still matters most.** The fixed global reference remains the worst localization variant (patch F1 5.77 %, IoU 0.08 %). The per-fabric design is the single most important choice in the system.

6. **Finer refinement helps a little, and only where the coarse stage is right.** The cascade raises mean pixel IoU from 1.36 to 1.52 % ($64 \rightarrow 32$) and 1.55 % (32 only), and pooled IoU from 1.12 to 1.17 %. Pixel recall drops ($48.4 \rightarrow 43.8$ / 46.1 %), because a smaller window crosses its own τ less often.

   - **Wide defects:** broken picks improve from 10.9 to 11.8 % IoU.
   - **Small defects:** they stay at or below 0.03 % IoU (fuzzy ball, nep, warp ball). Finer windows cannot recover what the descriptor does not separate.

   The default stays at [64]. It was fixed before the experiment, and adopting a test-set winner would bias the reported numbers [4].

7. **The 25-pixel label rule hid the smallest defects.** Counting a window as defective when it holds *any* defect pixel changes the ground truth, not the detector. Under that rule:

   - Patch F1 rises from 18.38 to 19.77 % and window AUROC from 0.536 to 0.557.

- The five warp balls (11–24 px masks) become measurable for the first time. Their patch recall is 40 % and their window AUROC 0.826, so the descriptor *does* respond to them.

The frozen protocol keeps the 25-pixel rule for comparability with Weeks 5–6. The end-semester report will show both.

8. **The image-level threshold is robust.** At $\alpha = 0.10$ detection rises to 61.9 % (accuracy 69.9 %), still with no false alarm. The only variants with image-level false alarms are the One-Class SVM (4 of 28), LBP-only and strip-normalised scoring (1 of 28 each).
9. **Descriptors are task-dependent.** Gabor-only features give the best image-level ranking (AUROC 0.878, accuracy 72.9 %) but worse localization (IoU 0.89 %). For the end-semester system, the image decision and the pixel mask may use different blocks. This is a configuration change, not new code.

Testing and Reproducibility

The suite grew from 72 to 86 tests. New tests cover:

- the three-way split (sizes, disjointness, seeding, scarce-data rules)
- the fold assignment (every strip in exactly one fold, single-strip fabrics kept out, seeded)
- background selection (only zero-defect windows of the named training strips)
- the conformal p-values, including an empirical check of the α bound
- image-level metrics and defective-only summaries
- the cascade configuration (even, strictly coarse-to-fine sizes)
- the cascade geometry (a 32×32 window can only fire inside a flagged 64×64 window)

Every table of results in this report is produced by `reports/week-7/make_tables.py` from the committed CSVs. Each result folder carries its `run_config.json` (git SHA, full configuration, package versions).

Commit History

Table 5: Commits of the week on `feat/week-7`, oldest first (one-line messages).

Hash	Message
`df63041`	feat: add image-level detection with conformal confidence, in-domain cross-validated reference and refinement cascade
`7e2bd6e`	feat: add in-domain, cascade, label-rule and image-alpha variants to the ablation grid
`5f84aaf`	chore: publish week 7 AFID benchmark and ablation results
`5f4e709`	docs: update README for the week 7 protocol, image-level detection and results
—	docs: add week 7 progress report (this document, with its table generator)

Literature Review

Topic / Paper	Key Takeaway	Relevance to the Project
Algorithmic Learning in a Random World [1]	Conformal prediction turns any nonconformity score into p-values with finite-sample validity under exchangeability.	Basis of the image-level confidence (Eq. 3) and its false-alarm guarantee.
Inductive conformal anomaly detection [2]	Anomaly detection with a held-out calibration set; the false-alarm rate is bounded by the significance level.	Exactly our setting: a defect-free calibration set, a strip-level score, and a significance level α.
Dataset Shift in Machine Learning [3]	Train and test distributions may differ in their inputs while the labelling rule stays the same.	The AFID capture shift, and why in-domain background was tried.
Cross-validation pitfalls [4]	Choosing a model or threshold using the evaluation data biases performance estimates upward.	Strip-level folds, never window-level, and no threshold tuned on test strips.
The relationship between precision-recall and ROC curves [5]	ROC AUC is threshold-free and insensitive to class balance.	Per-strip window AUROC and image-level AUROC as threshold-free diagnostics.
Summed-area tables [6]	$O(1)$ rectangle sums.	Window defect-pixel counts for labels, background selection and cascade containment.
AFID [7]	7 fabrics, 12 defect classes, defect-free and defective strips with masks.	Dataset; source of the protocol's 105 + 28 test strips.

Challenges Faced

- **Leakage risk in the in-domain reference.** Using defect-free windows of defective strips is only valid if a strip's own windows never reach its model. A window-level split would have leaked through overlapping windows and shared acquisition conditions, so the folds are by strip, with a second strip-level split for calibration inside each fold.
- **Small calibration sets.** With 4 calibration strips per fabric, per-fabric conformal p-values could not go below 0.2. Normalising image scores by τ_f (Eq. 2) lets one pooled set of 28 strips calibrate all fabrics.
- **Cost of the 32×32 level.** It brought about $15\times$ more windows than the 256×256 level. The per-image, per-scale feature cache made it a one-off cost.
- **Telling the method apart from the data.** Image-level AUROC alone would have suggested strong detection. Breaking it down by fabric, and adding the window AUROC, showed how much of it comes from the capture shift.

Plan for Next Week (Week 8)

- **Defect-aware threshold for the in-domain model.** Its ranking is the best we have (window AUROC 0.609). Set its τ by nested, strip-level validation on defect masks of training folds only, instead of $\mu + k\sigma$ on background windows.
- **Report image-level results on shift-free fabrics separately** (03, 04, 06) in the end-semester report, so the capture shift cannot inflate them.
- **End-semester report and presentation.** Use the frozen protocol, final tables regenerated by script, inspection figures, and the honest limitations: window separability, the capture shift and small defects.

Conclusion

Week 7 completed the planned items:

- a three-way protocol with true negatives
- image-level detection with a conformal confidence: 68.4 % accuracy, AUROC 0.790, no false alarms on 28 held-out defect-free strips at $\alpha = 0.05$
- a cross-validated in-domain reference
- a refinement cascade and the small-defect protocol study

The new threshold-free window AUROC changes how the remaining gap should be read. The default model ranks defect windows above background only slightly better than chance (0.536). In-domain background improves that ranking (0.609) but not the thresholded metrics. So the next gains will come from defect-aware threshold selection and more separable fine-scale descriptors, not from further tuning of k or the scorer.

References

[1] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.

[2] R. Laxhammar and G. Falkman, "Inductive conformal anomaly detection for sequential detection of anomalous sub-trajectories," *Ann. Math. Artif. Intell.*, vol. 74, no. 1–2, pp. 67–94, 2015.

[3] J. Quiñonero-Candela, M. Sugiyama, A. Schwaighofer, and N. D. Lawrence, Eds., *Dataset Shift in Machine Learning*. Cambridge, MA, USA: MIT Press, 2009.

[4] S. Varma and R. Simon, "Bias in error estimation when using cross-validation for model selection," *BMC Bioinformatics*, vol. 7, art. 91, 2006.

[5] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proc. 23rd Int. Conf. Mach. Learn. (ICML)*, 2006, pp. 233–240.

[6] F. C. Crow, "Summed-area tables for texture mapping," *Comput. Graph. (Proc. SIGGRAPH)*, vol. 18, no. 3, pp. 207–212, 1984.

[7] J. Silvestre-Blanes, T. Albero-Albero, I. Miralles, R. Pérez-Llorens, and J. Moreno, "A public fabric database for defect detection methods and results," *AUTEX Res. J.*, vol. 19, no. 4, pp. 363–374, Dec. 2019, doi: 10.2478/aut-2019-0035.